WIPER MOTOR INSPECTION/ GEAR COVER REPLACEMENT

AFFECTED VEHICLES: **2000-2001 MODEL YEAR**

TAURUS, SABLE, CONTINENTAL, TOWN CAR, FOCUS, EXPEDITION, NAVIGATOR, F-150, EXCURSION AND F-250 THROUGH F-750

AFFECTED WIPER MOTORS:

VEHICLE	WIPER MOTOR BUILD DATES AFFECTED
Taurus, Sable, Continental, Town Car, Focus, Expedition, Navigator, F-150, Excursion, F-250 through F-750	02/10/00 - 06/26/00

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OVERVIEW

This procedure gives the detail for the inspection of the wiper motor and the installation of the wiper motor gear cover. For all applications, the wiper motor will require an inspection of the build date code information to determine if a ***new*** gear cover must be installed. Due to variations in wiper systems, the affected vehicles will require different levels of disassembly to access the build date code information and to install a ***new*** cover.

I. PRE-REPAIR INSPECTION

1. Before performing any repairs, verify that the wiper system is operational on low and high-speed settings, then turn off the wiper system and ensure that the wiper blades travel to the park position.
2. If the motor does not function, perform standard pinpoint diagnostic procedures and repair the system accordingly.

   CAUTION: Wiper system must be parked prior to completing the gear cover replacement procedure to ensure that the internal gear will be in the correct position.

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II. DETERMINE IF WIPER MOTOR GEAR COVER MUST BE REPLACED

F-150, EXPEDITION, NAVIGATOR, CONTINENTAL, TAURUS, SABLE and FOCUS

1. Remove the cowl panel grille. Refer to the service procedure in Section 501-02 of the appropriate Workshop Manual.
2. Locate the wiper motor build date. See Figures 1 and 2. The build date is etched into the outer case of the wiper motor. An adhesive label may be covering the information. This label does not contain the proper information and must be peeled back to read the build code information.
3. Use the motor build date to determine if gear cover replacement is required:

MOTOR BUILD DATE	ACTION REQUIRED
02/10/00 - 06/26/00 (through and including)	Replace motor gear cover. Proceed to Section III and perform Wiper Motor Gear Cover Replacement procedure.
Before 02/10/00 or after 06/26/00	No repair is required. Install the cowl panel grille. Release vehicle to customer.
Not present or cannot be read	Replace motor gear cover. Proceed to Section III and perform Wiper Motor Gear Cover Replacement procedure.

EXCURSION, F-250 THROUGH F-750 and TOWN CAR

1. Remove the wiper module (motor and linkage assembly) from the vehicle and place it on a bench with the back of the wiper motor facing upward. Refer to the service procedures in Section 501-16 of the appropriate Workshop Manual.
2. Locate the wiper motor build date. See Figures 1 and 2. The build date is etched into the outer case of the wiper motor. An adhesive label may be covering the information. This label does not contain the proper information and must be peeled back to read the build code information.
3. Use the motor build date to determine if gear cover replacement is required:

MOTOR BUILD DATE	ACTION REQUIRED
02/10/00 - 06/26/00 (through and including)	Replace motor gear cover. Proceed to Section III and perform Wiper Motor Gear Cover Replacement procedure.
Before 02/10/00 or after 06/26/00	No repair is required. Install the wiper module and cowl panel grille. Verify wiper system function and release vehicle to customer.
Not present or cannot be read	Replace motor gear cover. Proceed to Section III and perform Wiper Motor Gear Cover Replacement procedure.

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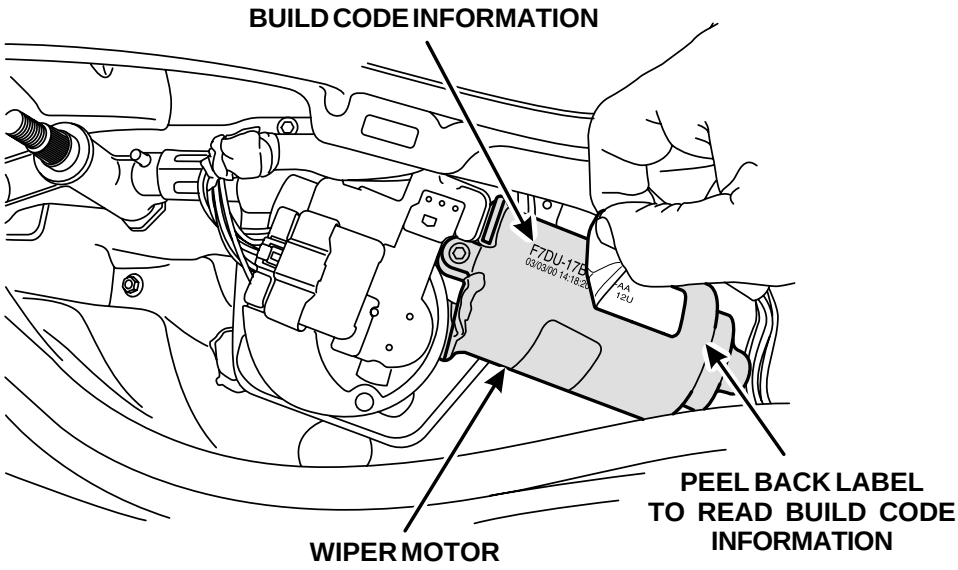


FIGURE 1

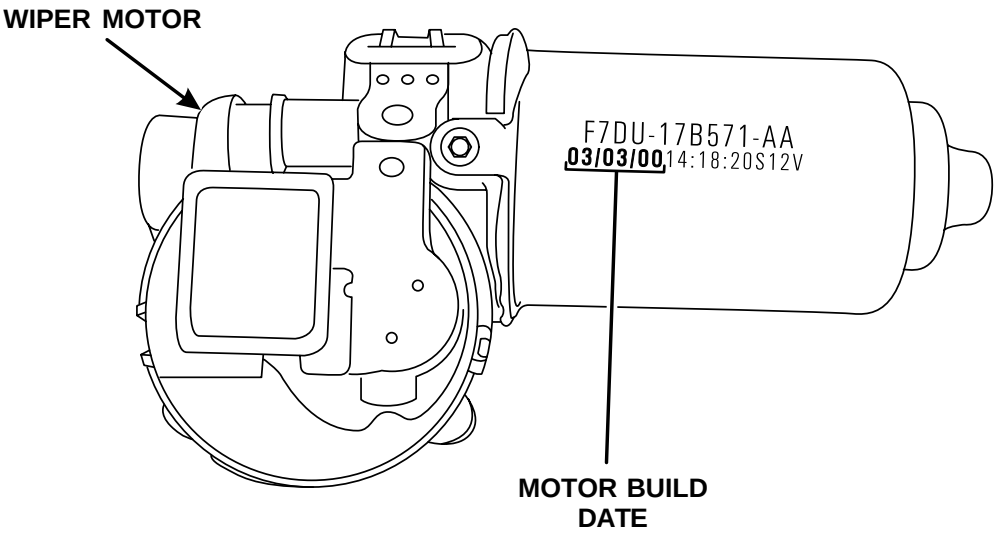


FIGURE 2

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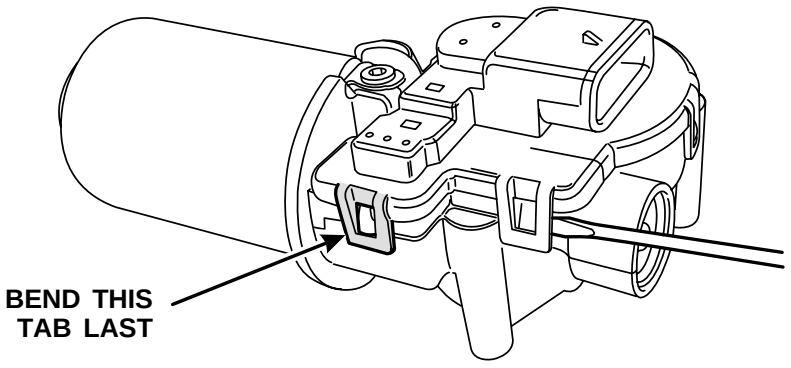
III. WIPER MOTOR GEAR COVER REPLACEMENT

REMOVE WIPER MOTOR GEAR COVER

VEHICLE	ACTION REQUIRED
F-150, Expedition, Navigator, Continental, Taurus and Sable	The gear cover can be replaced without removing the wiper module (motor and linkage assembly) from the vehicle.
F-250 through F-750, Excursion and Town Car	The gear cover must be replaced with the wiper module (motor and linkage assembly) removed from the vehicle. The module was removed during the inspection on these vehicles.
Focus	The gear cover must be replaced with the wiper module (motor and linkage assembly) removed from the vehicle. Remove the wiper module and place it on a bench with the back of the wiper motor facing upwards. Refer to the service procedure in Section 501-16 of the appropriate Workshop Manual.

1. Using a screwdriver, bend the five gear cover plastic lock tabs away from the motor housing. See Figure 3.
2. **CAUTION: The gear cover must be squarely lifted off of the motor housing. Any rocking or sideways motion may bend the three internal electrical connector clips. Pry up around the perimeter of the gear cover to break the sealant from the cover.**

   Once the sealant has been broken loose, squarely lift the gear cover off of the motor housing. Discard the gear cover.
3. Inspect the inside of the wiper motor housing for corrosion.
   - If there are no signs of corrosion inside the housing, continue with this procedure.
   - If the inside of the housing is corroded, water has entered the wiper motor. Contact the Special Service Support Center at 1-800-325-5621 for further instruction.



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FIGURE 3

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REMOVE SEALANT

CAUTION: Do not use solvents or compressed air to remove the sealant from the motor gear housing. Any solvent entering the gear cavity may damage the motor. Also, remove all large pieces of sealant that enter the gear housing cavity during cleaning. Small particles of sealant left in the cavity are acceptable.

NOTE: It is not necessary to remove the existing sealant attached to the vertical surfaces inside the gear housing.

1. Remove the existing sealant from the top surface of the gear housing and around each of the five mounting lug areas. To avoid damaging the motor, extra care must be taken when removing sealant from the area around the internal electrical connector clips. See Figure 4.

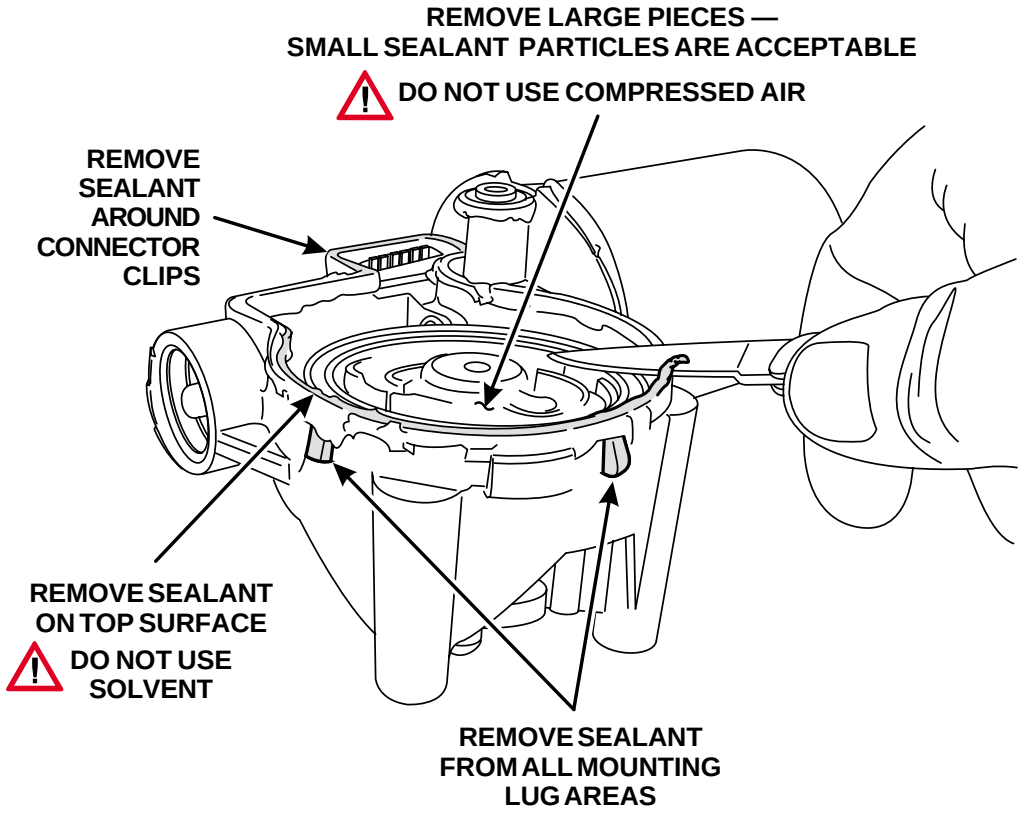


FIGURE 4

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INSPECT MOTOR ELECTRICAL CONNECTOR CLIPS

1. Inspect the three (3) internal electrical connector clips. **All three clips MUST be aligned against the plastic terminal holder.** If any of the clips are misaligned or touching the motor housing, they must be straightened and positioned as shown in the illustration. See Figure 5.

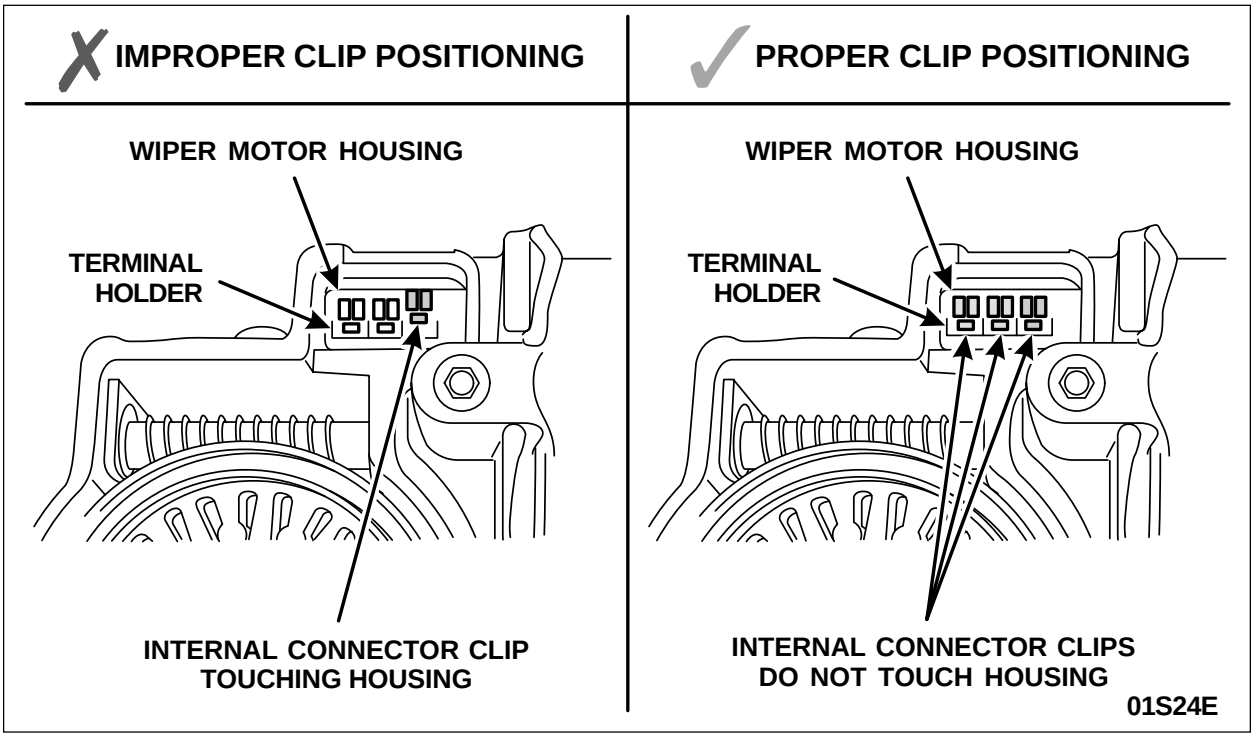


FIGURE 5

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PREPARE REPLACEMENT COVER

1. **IMPORTANT: Using your thumb, bend locking tab "E" approximately 60-90 degrees inward (see Figure 6) then bend it back approximately to the position illustrated in Figure 7. This will improve the engagement of the locking tab with the motor gear housing.**

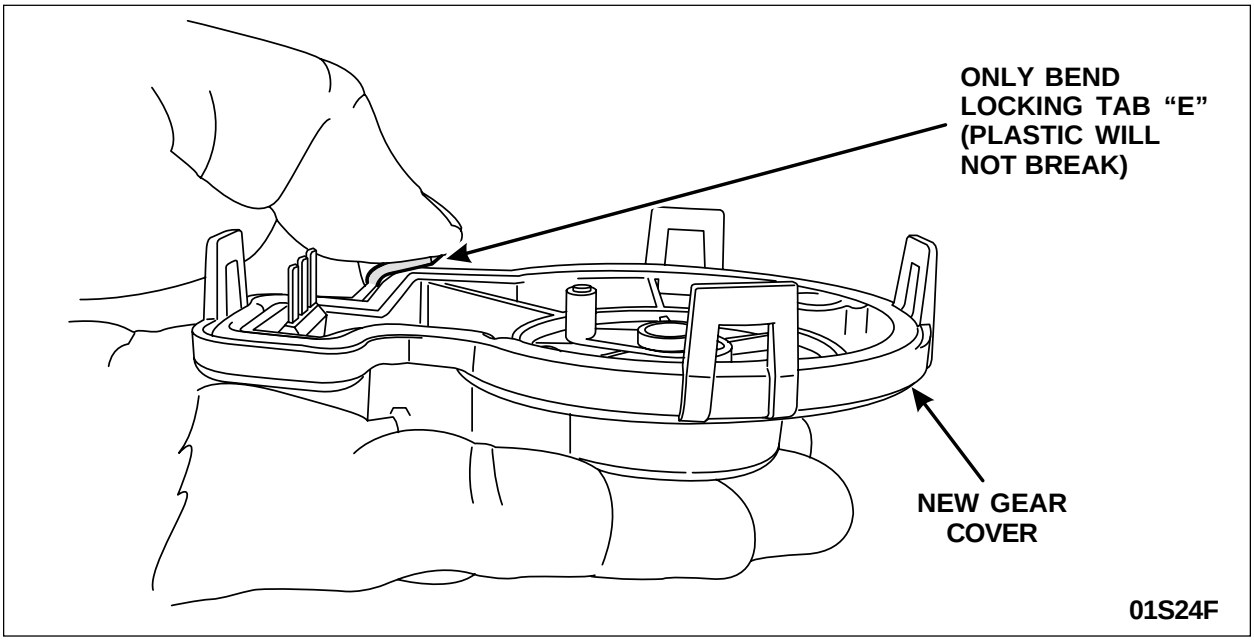


FIGURE 6

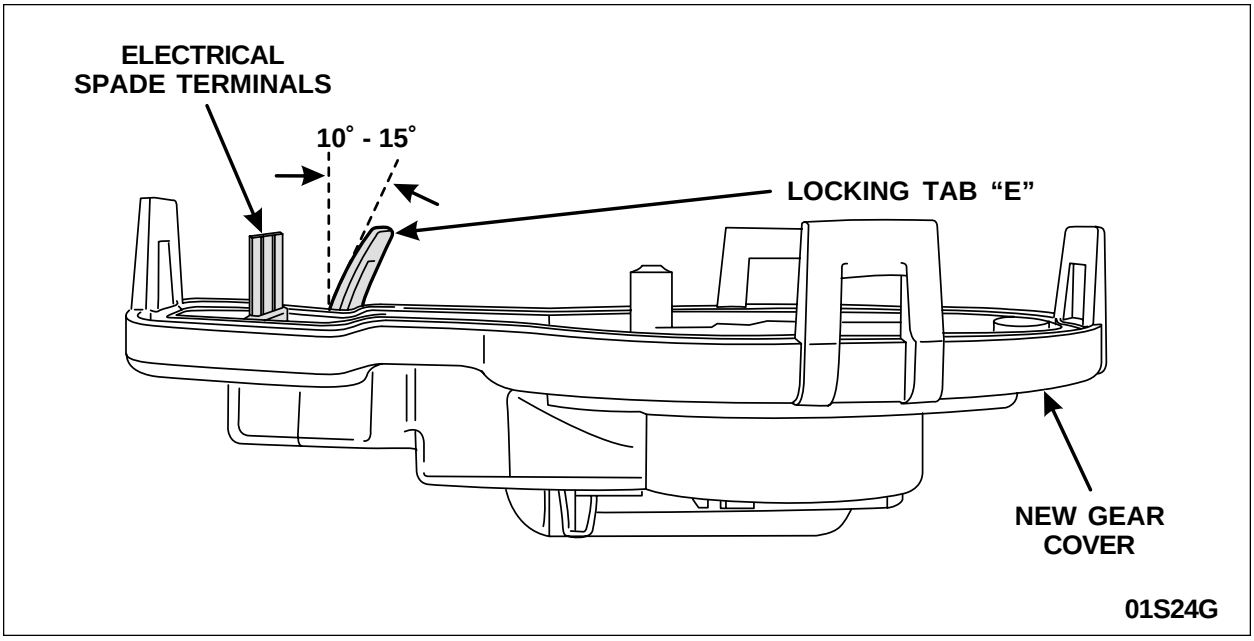


FIGURE 7

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2. Verify that the three (3) electrical spade terminals are not bent. If necessary, straighten them so that they are perpendicular to the cover as illustrated in Figure 8.
3. Make sure that the foam gasket is properly seated in the ***new*** cover as shown in Figure 8.
4. **Make sure that the white cam follower arm is positioned as shown in Figure 8.**

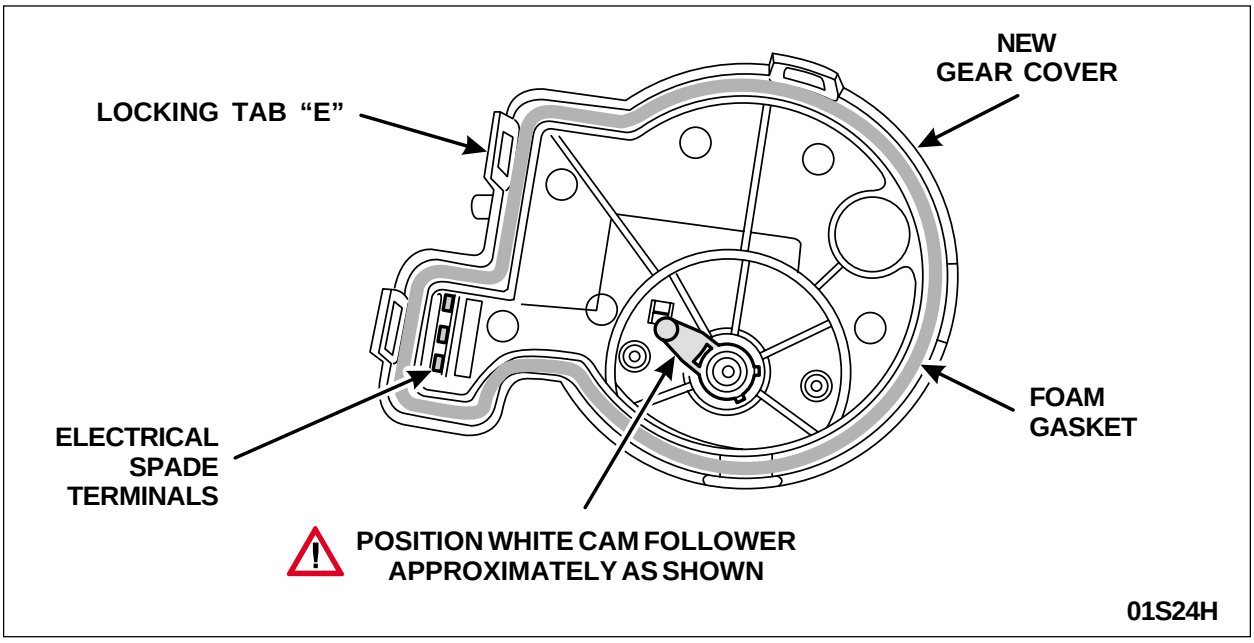


FIGURE 8

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INSTALL REPLACEMENT COVER

1. **IMPORTANT: Position cover on gear housing so that each plastic locking tab is outside of the gear housing cavity and is resting over the mounting lugs. Ensure that the electrical spade terminals are lined up with the connector clips. See Figure 9.**
2. Using two hands, push the gear cover downward onto the gear housing uniformly, to ensure that the electrical terminals remain in alignment. Push down firmly until all five locking tabs are latched correctly. See Figure 10.
3. **CAUTION: If any of the locking tabs cannot be correctly latched as shown in Figure 11, replace the complete motor assembly. DO NOT attempt to remove the cover and reinstall it.**

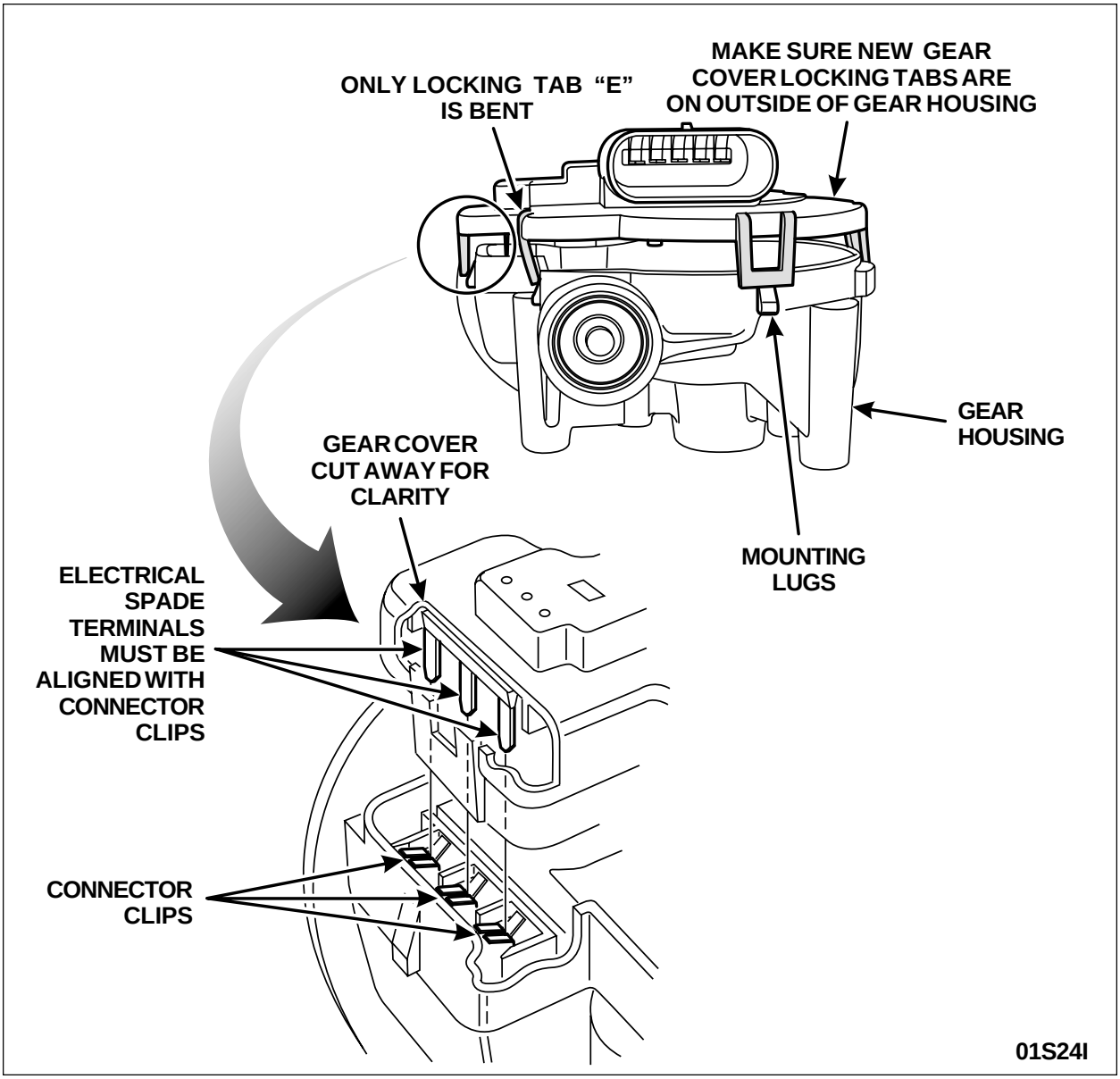


FIGURE 9

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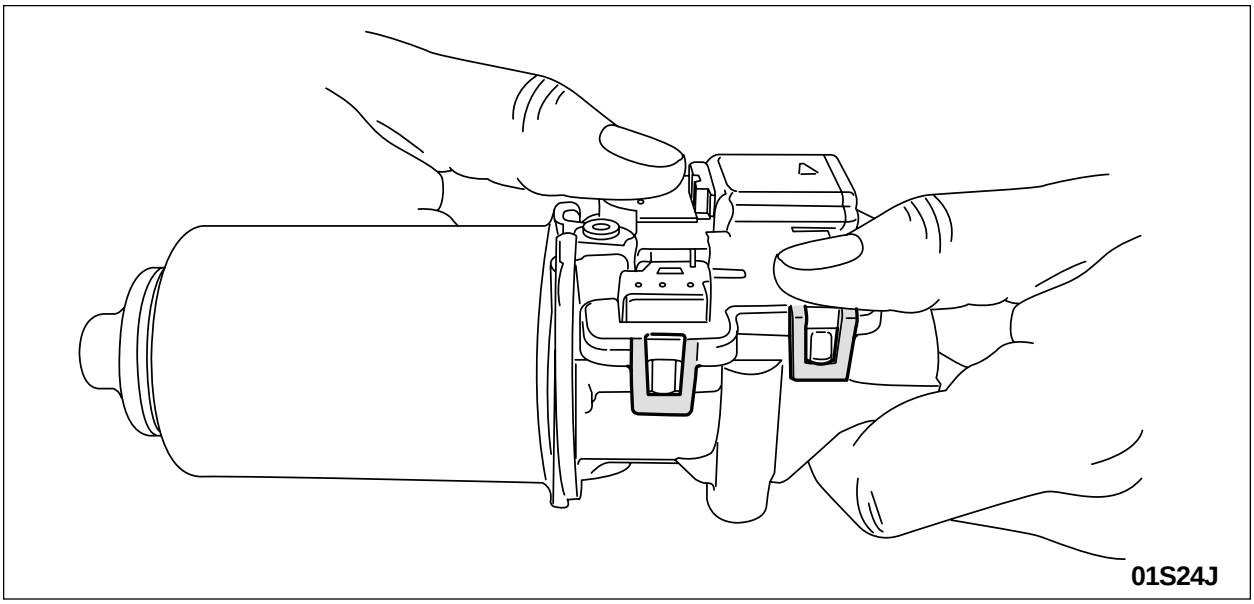


FIGURE 10

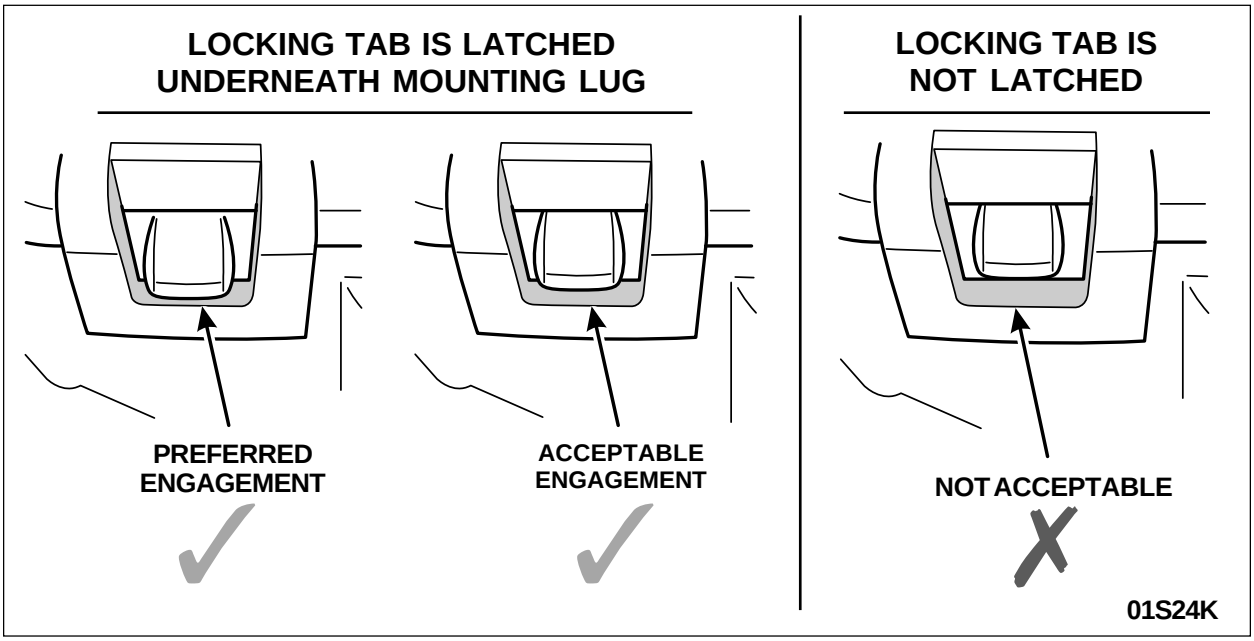


FIGURE 11

IV. VERIFY WIPER MOTOR OPERATION

1. If removed, install the wiper module and/or the cowl panel grille. Refer to the procedures in Sections 501-02 and 501-16 of the appropriate Workshop Manual.
2. Check the operation of the wiper motor (high speed, low speed and park).

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